RAMCO VIDYA MANDIR SR. SEC. SCHOOL, THAMARAIKULAM

CLASS :	**TIME : 3 Hrs**
EXAM NO :	**MARKS : 80**

SECTION A (1 mark each) (14 × 1 = 14)

1) In Ctenophores, which of the following features is NOT correct?
a) They possess tissue level of organization
b) They have eight rows of ciliated comb plates
c) They exhibit only extracellular digestion
d) They exhibit bioluminescence
2) The characteristic feature that distinguishes Ctenophores from Cnidarians is:
a) Radial symmetry
b) Absence of cnidoblasts
c) Diploblastic organization
d) Intracellular digestion
3) Assertion (A): Ctenophores are commonly known as sea walnuts or comb jellies.
Reason (R): They possess eight external rows of ciliated comb plates which help in locomotion.
a) Both A and R are correct and R is the correct explanation of A
b) Both A and R are correct but R is not the correct explanation of A
c) A is correct but R is incorrect
d) A is incorrect but R is correct
4) Which of the following statements about Ctenophores is NOT true?
a) They are exclusively marine organisms
b) They exhibit radial symmetry
c) They reproduce sexually with separate sexes
d) They exhibit both intracellular and extracellular digestion
5) In a leaf, the pulvinus refers to:
a) A specialized sheath covering the stem
b) A swollen leaf base found in some leguminous plants
c) The flattened green portion of the leaf
d) The stalk that connects the lamina to the stem
6) Case Study: Examine the floral diagram shown in a biology textbook.
Which of the following floral characters cannot be determined from a floral diagram?
a) Position of the mother axis
b) Number of floral parts
c) Aestivation of sepals and petals
d) Time of anthesis
7) In a typical flower, if the ovary is situated above the attachment of other floral parts on the thalamus, the flower is described as:
a) Epigynous
b) Perigynous

c) Hypogynous
d) Actinomorphic
8) Which of the following is the correct floral formula for a flower that is bisexual, actinomorphic with 5 united sepals, 5 free petals, 10 free stamens and superior ovary with 5 united carpels?
a) ⊕ K(5) C5 A10 G(5)
b) ⊕ K(5) C5 A10 G5
c) % K(5) C5 A10 G(5)
d) ⊕ K5 C(5) A10 G(5)
9) Assertion (A): In vexillary aestivation, the largest petal overlaps the two lateral petals which in turn overlap the two smallest anterior petals.
Reason (R): Vexillary aestivation is characteristic of the flowers belonging to family Fabaceae.
a) Both A and R are correct and R is the correct explanation of A
b) Both A and R are correct but R is not the correct explanation of A
c) A is correct but R is incorrect
d) A is incorrect but R is correct
10) Which of the following statements about phyllotaxy is INCORRECT?
a) It refers to the pattern of arrangement of leaves on the stem
b) In alternate phyllotaxy, a single leaf arises at each node
c) In opposite phyllotaxy, leaves at a node lie perpendicular to leaves at the next node
d) Whorled phyllotaxy is seen when more than two leaves arise at a node
11) In the region of maturation of the root:
a) Cells divide rapidly to increase the length of the root
b) Cells elongate to increase the length of the root
c) Root hairs develop from epidermal cells
d) Root cap cells are formed to protect the meristematic region
12) When the incisions of the lamina reach up to the midrib breaking it into a number of leaflets, the leaf is called:
a) Simple
b) Compound
c) Pinnate
d) Palmate
13) A monocotyledonous seed differs from a dicotyledonous seed in having:
a) Single cotyledon and endospermic nature
b) Two cotyledons and non-endospermic nature
c) Single cotyledon and non-endospermic nature
d) Aleurone layer around the plumule
14) In which type of placentation are the ovules borne on the inner wall of the ovary or on the peripheral part of the septa?
a) Axile
b) Parietal
c) Free central
d) Marginal

SECTION B (2 marks each) (10 × 2 = 20)

15) Explain how the initial emphasis on detailed description in biology, which seemed boring to students, eventually became meaningful in the development of reductionist biology.
16) Analyze the structural differences between the stomatal apparatus in dicots and monocots, and explain how these differences might affect their physiological functions.
17) Compare the organization of vascular bundles in monocotyledonous stems versus dicotyledonous stems, and explain the functional significance of these differences.
18) Explain how the structural features of bulliform cells in grass leaves contribute to drought adaptation, with reference to their turgidity and function.
19) Differentiate between the anatomical organization of the endodermis in roots compared to stems, and relate these differences to their respective physiological roles.
20) Analyze why secondary growth is possible in dicot stems but not in monocot stems, with reference to the arrangement and structure of their vascular bundles.
21) Explain how the structural differences between the adaxial and abaxial epidermis of a dorsiventral leaf are adaptations to their environmental exposure.
22) Compare the structural organization of the stele in dicot and monocot roots, and explain how these differences affect their physiological capabilities.
23) Analyze how the position and structure of sclerenchymatous tissues in stems contribute to mechanical support in both monocots and dicots.
24) Explain how the arrangement of palisade and spongy parenchyma in dorsiventral leaves represents an adaptation to optimize photosynthesis while minimizing water loss.

SECTION C (3 marks each) (7 × 3 = 21)

25) Compare and contrast the arrangement of vascular bundles in monocot and dicot stems, and explain how these structural differences relate to their functional adaptations in supporting the plant body.
26) Examine how the structure of dorsiventral (dicot) leaves demonstrates adaptation for efficient photosynthesis through the arrangement of palisade and spongy parenchyma. How do these structural features maximize light capture and gas exchange?
27) Analyze the significance of the endodermis in roots with reference to the Casparian strips. How does this specialized tissue layer contribute to selective absorption of minerals and water movement in plants?
28) Explain how the structural organization of bulliform cells in monocot leaves represents an adaptation to environmental stress conditions. What physiological mechanism allows these cells to regulate water loss?
29) Differentiate between open and closed vascular bundles in plants, and evaluate the evolutionary significance of cambium presence in relation to secondary growth potential in different plant groups.
30) Critically analyze how the structure of stomatal apparatus varies between monocots and dicots, and explain how these differences affect their efficiency in regulating transpiration under different environmental conditions.
31) Examine how the structural organization of ground tissue in stems contributes to both mechanical support and storage functions. How do these anatomical features vary between plants adapted to different habitats?

SECTION D (4 marks each) (2 × 4 = 8)

HARD Case-Based/Competency/Value-Based Questions (4 marks each)
Question 1: Investigating Ctenophore Bioluminescence
A marine biology research team collected several specimens of comb jellies (Ctenophora) from coastal waters. They observed that the specimens exhibited bright blue-green bioluminescence when disturbed in dark conditions. The team extracted the bioluminescent substance and analyzed it under controlled laboratory conditions. They found that the bioluminescence occurred through a chemical reaction requiring calcium ions, and the light was produced without generating heat. Further investigation revealed that the bioluminescent properties were more pronounced in deeper-water species compared to those found in shallow waters.
The researchers hypothesized that this bioluminescence might serve multiple functions in the marine environment. They conducted experiments placing the ctenophores in tanks with potential predators and observed the bioluminescent response.
Based on your understanding of ctenophores and marine adaptations:
a) Explain why bioluminescence might be an evolutionary advantage for ctenophores, considering their tissue level organization and marine habitat. (2 marks)
b) Unlike cnidarians which have nematocysts for defense, ctenophores lack such specialized cells. How might bioluminescence serve as an alternative survival mechanism for these organisms? Suggest two possible ecological functions of this phenomenon. (2 marks)
Question 2: Comparative Plant Anatomy Investigation
A botany student conducted an experiment to study the anatomical adaptations of plants to different environmental conditions. The student collected leaf samples from two different plant species growing in the same botanical garden - one from a plant growing in a shaded, moist area and another from a plant growing in a sunny, dry location.
Upon microscopic examination of the transverse sections of both leaf samples, the following observations were recorded:

Feature	Sample A	Sample B
Cuticle thickness	Very thin	Thick and waxy
Stomatal density	Higher on abaxial surface	Almost equal on both surfaces
Palisade parenchyma	Single layer	Multiple layers
Spongy parenchyma	Extensive with large air spaces	Compact with smaller air spaces
Bulliform cells	Absent	Present
Vascular bundles	Widely spaced	Closely packed

a) Based on the anatomical features observed, identify which sample (A or B) belongs to a plant adapted to dry conditions, and explain how two of these features specifically help the plant survive in water-limited environments. (2 marks)
b) Sample B shows characteristics typical of a particular group of flowering plants. Identify whether this sample likely belongs to a monocot or dicot plant, citing specific anatomical evidence from the table. How do these anatomical adaptations reflect the evolutionary history of this plant group? (2 marks)

SECTION E (5 marks each) (3 × 5 = 15)

HARD Long Answer (5 marks each) Questions with Internal Choice

Question 1

Describe in detail the phylum Ctenophora, explaining its distinctive features including body organization, locomotion method, bioluminescence, and reproductive characteristics. Compare and contrast Ctenophora with other diploblastic phyla, highlighting their evolutionary significance.

OR

Explain the tissue organization in plants with special reference to the three tissue systems (epidermal, ground, and vascular). Describe how these systems are structurally and functionally adapted in stems and roots of both monocotyledonous and dicotyledonous plants.

Question 2

Discuss the morphology of a dorsiventral leaf in detail, explaining its various parts and their functions. Describe how the internal anatomy of a dicotyledonous leaf correlates with its physiological functions, particularly focusing on the adaptations for photosynthesis and gas exchange.

OR

Elaborate on the reproductive system of frogs. Explain the anatomical structure of male and female reproductive organs, their functional significance, and the process of external fertilization. Discuss how the reproductive adaptations of frogs reflect their amphibious lifestyle.

Question 3

Compare and contrast the stem anatomy of monocotyledonous and dicotyledonous plants with labeled diagrams. Explain how their structural differences relate to their functional adaptations and evolutionary history. Discuss the significance of vascular bundle arrangement in these two groups.

OR

Describe the digestive and respiratory systems of frog in detail. Explain how these systems are adapted for the frog's dual lifestyle in aquatic and terrestrial environments. Discuss the evolutionary significance of these adaptations in the context of vertebrate evolution from aquatic to terrestrial habitats.

SUMMARY OF PAPER GENERATION

Section A: 14 questions | Chapters 4, 5 | Context size 27643 chars
Section B: 10 questions | Chapters 5, 6 | Context size 26654 chars
Section C: 7 questions | Chapters 6, 7 | Context size 27717 chars
Section D: 2 questions | Chapters 4, 5, 6, 7 | Context size 55361 chars
Section E: 3 questions | Chapters 4, 5, 6, 7 | Context size 55361 chars